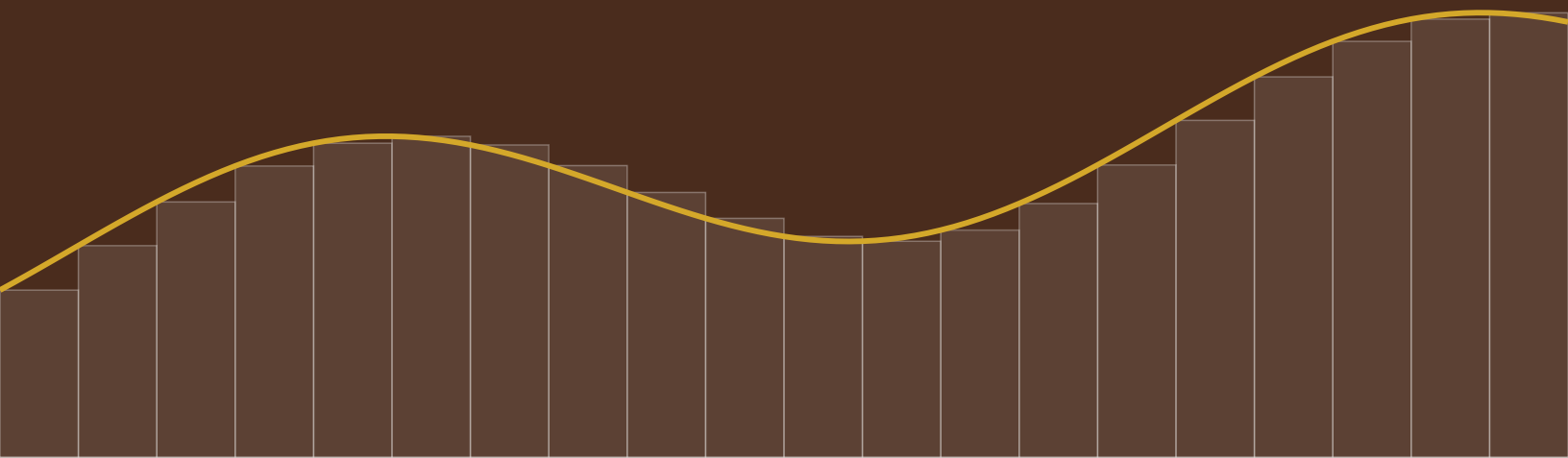




CARMEL CATHOLIC HIGH SCHOOL

AP Calculus BC

Course Notes

Based on Demana, Franklin D., et al. *Calculus: Graphical, Numerical, Algebraic*. 6th ed., AP ed., Pearson, 2020.

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Prerequisites for Calculus

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0.1 Linear Functions

Increments and Slope

Unlike algebra, calculus doesn't deal with exact values or inequalities. Instead it deals with changing quantities, rates of change, and Increments.

Definition 0.1.1 (*Increments*) The change in a variable is called an *increment* in that variable. Let x be a variable.

$$\Delta x = x_2 - x_1$$

Using increments, you can find the *rate of change* or slope of any *linear function*.

Definition 0.1.2 (*Slope*) Let (x_1, y_1) and (x_2, y_2) be two points on the graph of a linear function.

$$m = \frac{\Delta y}{\Delta x} \text{ or } \frac{y_2 - y_1}{x_2 - x_1}$$

Point-Slope Equation of a Linear Function

Other Linear Equation Forms

Parallel and Perpendicular Lines

Applications of Linear Functions

Solving Two Linear Equations Simultaneously

0.2 Functions and Graphs

Functions

Domains and Ranges

Viewing and Interpreting Graphs

Even Functions and Odd Functions – Symmetry

Piecewise-Defined Functions

Absolute Value Function

Composite Functions

0.3 Exponential Functions

Exponential Growth

Exponential Decay

Compound Interest

The Number e

0.4 Parametric Equations

Relations

Circles

Ellipses

Lines and Other Curves

0.5 Inverse Functions and Logarithms

One-to-One Functions

Inverses

Finding Inverses

Logarithmic Functions

Properties of Logarithms

Applications

0.6 Trigonometric Functions

Radian Measure

Graphs of Trigonometric Functions

Periodicity

Even and Odd Trigonometric Functions

Transformations of Trigonometric Graphs

Inverse Trigonometric Functions

Prologue – Foundations of Calculus

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P.1 Velocity and Distance

Finding Distance

Finding Velocity

CHAPTER 1

Limits and Continuity

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1.1 Rates of Change and Limits

Average and Instantaneous Velocity

Definition of Limit

Properties of Limits

One-Sided and Two-Sided Limits

Squeeze Theorem

1.2 Limits Involving Infinity

Finite Limits as $x \rightarrow \pm\infty$

Squeeze Theorem Revisited

Infinite Limits as $x \rightarrow a$

End Behavior Models

“Seeing” Limits as $x \rightarrow \pm\infty$

1.3 Continuity

Continuity at a Point

Continuous Functions

Algebraic Combinations

Composites

Intermediate Value Theorem for Continuous Functions

1.4 Rates of Change, Tangent Lines, and Sensitivity

Average Rates of Change

Tangent to a Curve

Slope of a Curve

Normal to a Curve

Velocity Revisited

Sensitivity

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Derivatives

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2.1 Derivative of a Function

Definition of Derivative

Notation

Relationships Between the Graphs of f and f'

Graphing the Derivative from Data

One-Sided Derivatives

2.2 Differentiability

How $f'(a)$ Might Fail to Exist

Differentiability Implies Local Linearity

Numerical Derivatives on a Calculator

Differentiability Implies Continuity

Intermediate Value Theorem for Derivatives

2.3 Rules for Differentiation

Positive Integer Powers, Multiples, Sums, and Differences

Products and Quotients

Negative Integer Powers of x

Second and Higher Order Derivatives

2.4 Velocity and Other Rates of Change

Instantaneous Rates of Change

Motion Along a Line

Sensitivity to Change

Derivatives in Economics

2.5 Derivatives of Trigonometric Functions

Derivative of the Sine Function

Derivative of the Cosine Function

Simple Harmonic Motion

Jerk

Derivatives of the Other Basic Trigonometric Functions

CHAPTER 3

More Derivatives

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3.1 Chain Rule

Derivative of a Composite Function

“Outside-Inside” Rule

Repeated Use of the Chain Rule

Slopes of Parametrized Curves

Power Chain Rule

3.2 Implicit Differentiation

Implicitly Defined Functions

Lenses, Tangents, and Normal Lines

Derivatives of Higher Order

Rational Powers of Differentiable Functions

3.3 Derivatives of Inverse Trigonometric Functions

Derivatives of Inverse Functions

Derivative of the Arcsine

Derivative of the Arctangent

Derivative of the Arcsecant

Derivatives of the Other Three

3.4 Derivatives of Exponential and Logarithmic Functions

Derivative of e^x

Derivative of a^x

Derivative of $\ln x$

Derivative of $\log_a x$

Power Rule for Arbitrary Real Powers

Applications of Derivatives

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4.1 Extreme Values of Functions

Absolute (Global) Extreme Values

Local (Relative) Extreme Values

Finding Extreme Values

4.2 Mean Value Theorem

Mean Value Theorem

Physical Interpretation

Increasing and Decreasing Functions

Other Consequences

4.3 Connecting f' and f'' with the Graph of f

First Derivative Test for Local Extrema

Concavity

Points of Inflection

Second Derivative Test for Local Extrema

Learning About Functions from Derivatives

4.4 Modeling and Optimization

A Strategy for Optimization

Examples from Mathematics

Examples from Business and Industry

Examples from Economics

Modeling Discrete Phenomena with Differentiable Functions

4.5 Linearization, Sensitivity, and Differentials

Linear Approximation

Differentials

Sensitivity Analysis

Absolute, Relative, and Percentage Change

Sensitivity to Change

Newton's Method

Newton's Method May Fail

4.6 Related Rates

Related Rate Equations

Solution Strategy

Simulating Related Motion

CHAPTER 5

The Definite Integral

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5.1 Estimating with Finite Sums

Accumulation Problems as Area

Rectangular Approximation Method (RAM)

Volume of a Sphere

Cardiac Output

5.2 Definite Integrals

Riemann Sums

Terminology and Notation of Integration

Definite Integral as an Accumulator Function

Definite Integral and Area

Constant Functions

Integrals on a Calculator

Discontinuous Integrable Functions

5.3 Definite Integrals and Antiderivatives

Properties of Definite Integrals

Average Value of a Function

Mean Value Theorem for Definite Integrals

Connecting Differential and Integral Calculus

5.4 Fundamental Theorem of Calculus

Fundamental Theorem, Antiderivative Part

Graphing the Accumulator Function $\int_a^x f(t) dt$

Fundamental Theorem, Evaluation Part

The Real Meaning of the Fundamental Theorem

Area Connection

Analyzing Antiderivatives Graphically

5.5 Trapezoidal Sums

Trapezoidal Approximations**Other Algorithms****Error Analysis**

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Differential Equations and Mathematical Modeling

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6.1 Slope Fields and Euler's Method

Differential Equations

Slope Fields

Euler's Method

6.2 Antidifferentiation by Substitution

Indefinite Integrals

Leibniz Notation and Antiderivatives

Substitution in Indefinite Integrals

Substitution in Definite Integrals

6.3 Antidifferentiation by Parts

Product Rule in Integral Form

Solving for the Unknown Integral

Tabular Integration

Inverse Trigonometric and Logarithmic Functions

6.4 Exponential Growth and Decay

Separable Differential Equations

Law of Exponential Change

Continuously Compounded Interest

Radioactivity

Modeling Growth with Other Bases

Newton's Law of Cooling

6.5 Logistic Growth

How Populations Grow

Partial Fractions

The Logistic Differential Equation

Logistic Growth Models

Applications of Definite Integrals

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7.1 Accumulation and Net Change

Linear Motion Revisited

General Strategy

Accumulation of a Rate over Time

Coming and Going

Net Change from Data

Density

Work

7.2 Areas in the Plane

Area Between Curves

Area Enclosed by Intersecting Curves

Boundaries with Changing Functions

Integrating with Respect to y

Saving Time with Geometry Formulas

7.3 Volumes

Volume as an Integral

Square Cross Sections

Circular Cross Sections

Cylindrical Shells

Other Cross Sections

7.4 Lengths of Curves

A Sine Wave

Length of a Smooth Curve

Vertical Tangents, Corners, and Cusps

7.5 Applications from Science and Statistics

Work Revisited

Fluid Force and Fluid Pressure

Normal Probabilities

Sequences, L'Hospital's Rule, and Improper Integrals

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8.1 Sequences

Defining a Sequence

Arithmetic and Geometric Sequences

Graphing a Sequence

Limit of a Sequence

8.2 L'Hospital's Rule

Indeterminate Form $0/0$

Indeterminate Forms ∞/∞ , $\infty \cdot 0$, $\infty - \infty$

Indeterminate Forms 1^∞ , 0^0 , ∞^0

8.3 Relative Rates of Growth

Comparing Rates of Growth

Using L'Hospital's Rule to Compare Growth Rates

Sequential versus Binary Search

8.4 Improper Integrals

Infinite Limits of Integration

Integrands with Infinite Discontinuities

Test for Convergence and Divergence

Applications

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Infinite Series

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9.1 Power Series

Geometric Series

Representing Functions by Series

Differentiation and Integration

Identifying a Series

9.2 Taylor Series

Constructing a Series

Series for $\sin x$ and $\cos x$

Beauty Bare

Maclaurin and Taylor Series

Combining Taylor Series

Table of Maclaurin Series

9.3 Taylor's Theorem

Taylor Polynomials

The Remainder

Bounding the Remainder

Analyzing Truncation Error

Euler's Formula

9.4 Radius of Convergence

Convergence

n th-Term Test

Comparing Nonnegative Series

Ratio Test

Endpoint Convergence

9.5 Testing Convergence at Endpoints

Integral Test**Harmonic Series and p -series****Comparison Tests****Alternating Series****Absolute and Conditional Convergence****Intervals of Convergence****A Word of Caution**

Parametric, Vector, and Polar Functions

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10.1 Parametric Functions

Parametric Curves in the Plane

Slope and Concavity

Arc Length

Cycloids

10.2 Vectors in the Plane

Two-Dimensional Vectors**Vector Operations****Modeling Planar Motion****Velocity, Acceleration, and Speed****Displacement and Distance Traveled****10.3 Polar Functions****Polar Coordinates****Polar Curves****Slopes of Polar Curves****Areas Enclosed by Polar Curves****Spiral of Archimedes**